Haoming Ma, PhD

Research Scientist and Principal Investigator

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EDUCATION

Ph.D.	Department of Chemical & Petroleum Engineering, University of Calgary, Canada, 2023 Dissertation: *Data-driven Carbon Dioxide Enhanced Oil Recovery Models and Their Applications* Committee: Zhangxin Chen (chair), Sean T. McCoy (co-chair), Joule A. Bergerson, Christopher Clarkson, Steven Bryant, Xiaolong Yin (External)
M.Sc.	Computer Information and Technology, School of Engineering and Applied Science, University of Pennsylvania, Pennsylvania, 2025
M.Sc.	Energy Management and Policy, Department of Energy and Mineral Engineering, Pennsylvania State University, Pennsylvania, 2019
B. Sc.	Energy Engineering, Department of Energy and Mineral Engineering, Pennsylvania State University, Pennsylvania, 2018

PROFESSIONAL APPOINTMENTS

2025.03 to Present	***Staff Research Scientist and Principal Investigator***, Hydrogen Energy Research Center, School of Energy Resources, University of Wyoming
2025.03 to Present	***Affiliated Researcher,*** Center for Energy and Environmental Systems Analysis, University of Texas at Austin
2023.11 to 2025.03	***Post-doctoral Fellow***, Energy Emissions Modeling and Data Lab, Hildebrand Department of Petroleum and Geosystems Engineering, University of Texas at Austin
2023.09 to 2023.11	***Research Associate***, Reservoir Simulation Group, Department of Chemical & Petroleum Engineering, University of Calgary
2022.09 to 2023.05	***Sessional Instructor***, Reservoir Simulation Group, Department of Chemical & Petroleum Engineering, University of Calgary

RESEARCH INTERESTS

My research interest lies in developing Geo-Energy System Modeling and Analysis toolkits powered by machine-learning algorithms and multiscale remote-sensing data (e.g., from satellites to aircraft to ground-based sensors), coupling numerical methods with techno-economic analysis (TEA) and life cycle assessment (LCA) to enable the industrial-scale deployment of sustainable geo-energy extraction. Through this interdisciplinary effort, my work delivers technical, socioeconomic, and policy insights that advance a secure, affordable, and sustainable energy future.

I am particularly interested in the following topics:

- Geological CO_2 utilization associated with sequestration
- Methane emissions across geo-energy supply chains
- Subsurface hydrogen resources including extraction and storage
- Enhanced geothermal systems

AWARDS

2024	Best Paper Award, Frontiers in Energy
2023	SPE Canadian Educational Foundation General Scholarship, Society of Petroleum Engineers Educational Foundation
2022	Alberta Graduate Excellence Scholarship
2022	Canada First Research Excellence Fund Conference Grant
2022, 2023	Chemical & Petroleum Engineering Graduate Excellence Award, University of Calgary
2020-2022	International Graduate Tuition Award, University of Calgary
2019	Graduate Teaching Assistant Fellowship, Penn State University
2018	Outstanding Graduate Teaching Assistant Award, Penn State University
2016-2018	Leone Family Undergraduate Scholarship, Penn State University
2017	John and Elizabeth Holmes Teas Scholarship, Penn State University

Media Coverage

2024	Estimating emissions potential of decommissioned gas wells from shale samples, Research, Penn State News (April 22, 2024)
2023	Indigenous pathways program makes a difference for Schulich students, UToday, UCalgary News (March 30, 2023)

RESEARCH EXPERIENCE

2025-Present **Staff Research Scientist and Principal Investigator**, Hydrogen Energy Research Center, School of Energy Resources, University of Wyoming

Lead a research program that performs techno-economic and life-cycle analyses of innovative hydrogen-production and storage technologies, advancing Wyoming's energy-driven economy. Directly involved and secured **~$21M** research grant from U.S. Department of Energy (DOE) Office of Fossil Energy and Carbon Management (FECM) and National Science Foundation (NSF).

2023-2025 **Post-doctoral Fellow,** Energy Emissions Modeling and Data Lab, Hildebrand Department of Petroleum and Geosystems Engineering, University of Texas at Austin

Developed a measurement-informed life-cycle assessment (LCA) framework for global LNG supply chains and blue-hydrogen technologies. Contributed to the preparation of **$40M** proposals to the U.S. DOE FECM and industrial partners.

2023-2023 **Research Associate,** Reservoir Simulation Group, Department of Chemical & Petroleum Engineering, University of Calgary

2020-2023 **Research Assistant,** Reservoir Simulation Group, Department of Chemical & Petroleum Engineering, University of Calgary

Oversaw research team in reservoir modeling of geological CO_2 sequestration, geothermal energy recovery, and underground hydrogen storage; Developed artificial intelligence-based geosystem analysis tools to quantify trade-offs of CO_2 sequestration techniques; Led 5 research projects; Contributed to grant proposals; Mentored 2 PhD and 1 M.Sc. Funded by Natural Sciences and Engineering Research Council of Canada (NSERC), Canada First Research Excellence Fund (CFREF), and industrial partners.

TEACHING EXPERIENCE

Sole Instructor, University of Calgary

FA2022 Quantitative Engineering Foundations I

W2023 Quantitative Engineering Foundations II

Teaching Assistant, University of Calgary

FA2022 Reservoir Simulation

W2022 Life Cycle Assessment

FA2021 Reservoir Simulation

Teaching Assistant, Penn State University

SP2019	Risk Management in Energy Market
SU2018	Introduction to the Electricity Market
FA2018	Corporate Finance in Energy Market
FA2018	Fluid Mechanics
SP2019	Wind & Hydrology Power System
FA2017	Corporate Finance in Energy Market
FA2017	Fluid Mechanics

RESEARCH CONTRIBUTIONS

Peer-Reviewed Journal Articles (T=18; First-authored/Correspondence = 10)

Note: * denotes corresponding authors, underlined names are mentees

2025 [J17] Gao, X., Zhang, K.*, Ji, D., **Ma, H.***, Meng, Y., Jiang, S., Tian, W., Tang, Y., He, L., Liu, C., Lau, H. C., Chen Z., A proposed layout of CO_2 capture utilization and storage for coal-fired power plants in China. *Fuel*. 401: 135723.

2025 [J16] Di, C., Wei, Y., **Ma, H.**, Deng, P., Liu, B., Wang, K., Chen, Z.*.Accelerating geological gas storage simulations: A novel and efficient gases–brine phase-equilibrium calculation algorithm. *Renewable Energy*. 252: 123545

2025 [J15] Xue, Z., Wei, Z., **Ma ,H.**, Sun, Z., Lu, C., Chen, Z.*. Exploring the role of fracture networks in enhanced geothermal systems: Insights from integrated thermal-hydraulic-mechanical-chemical and wellbore dynamics simulations. *Renewable and Sustainable Energy Reviews*.215: 115636

2025 [J14] <u>Deng, P.</u>, **Ma, H.**, Song, J., Peng, X., Zhu, S., Xue, D., Jiang, L., Chen, Z.*, Carbon dioxide as cushion gas for large-scale underground hydrogen storage: Mechanisms and implications. *Applied Energy*. 388:125622

2025 [J13] **Ma, H.***, McCoy, S. T., Chen, Z.*, Development and comparison of reduced-order models for CO_2-enhanced oil recovery predictions, *Energy*, 320: 135313

2025 [J12] **Ma, H.***, Yang, Y., Xue, Z., Clarkson, C. R., Chen, Z.*, Transitioning from emission source to sink: Economic and environmental trade-offs for CO_2-enhanced coalbed methane recovery of Mannville coal Canada, *Science of the Total Environment,* 966: 178721

2024 [J11] <u>Xue, Z.</u>, <u>Zhang, Y.</u>, **Ma, H.***, Lu. Y., Zhang, K., Wei, Y., Wang, M., Wu, W., Chai, M., Sun, Z., Chen, Z.*, A combined neural network forecasting approach for CO_2 enhanced shale gas recovery. *SPE Journal*, 29 (08): 4459–4470. Paper Number: SPE-219774-PA

2024 [J10] Xue, Z., **Ma, H.***, Sun, Z., Lu, C., Chen, Z.*, Technical analysis of a novel economically mixed CO_2-water enhanced geothermal system. *Journal of Cleaner Production*, 448: 141749.

2024 [J9] Xue, Z., **Ma, H.**, Wei, Y., Wu, W., Sun, Z., Chai, M., Zhang, C., & Chen, Z., Integrated technological and economic feasibility comparisons of enhanced geothermal systems associated with carbon storage. *Applied Energy*, 359: 122757.

2024 [J8] Yang, Y., Liu, S.*, **Ma, H.**, Impact of unrecovered gas reserve on methane emissions from abandoned shale gas wells. *Science of the Total Environment*, 913: 169750.

2023 [J7] Deng, P., Chen, Z.*, Peng, X., Wang, J., Zhu, S., **Ma, H.**, Wu, Z., Optimized lower pressure limit for condensate underground gas storage using a dynamic pseudo-component model. *Energy*, 285: 129505.

2023 [J6] Xue, Z., Zhang, K., Zhang, C., **Ma, H.**, Chen, Z.*, Comparative data-driven enhanced geothermal systems forecasting models: a case study of Qiabuqia field in China. *Energy*, 280: 128255.

2023 [J5] Xue, Z., Yao, S., **Ma, H.**, Zhang, C., Zhang, K., Chen, Z.*, Thermo-economic optimization of an enhanced geothermal system (EGS) based on machine learning and differential evolution algorithms. *Fuel*, 340: 127569.

2023 [J4] **Ma, H.**, Sun, Z., Xue, Z., Zhang, C., Chen, Z.*, A systemic review of hydrogen supply chain in energy transition. *Frontiers in Energy*, 17: 102-122.

2022 [J3] **Ma, H.**, Yang, Y.*, Zhang, Y., Li, Z., Zhang, K., Xue, Z., Zhan J., Chen, Z.*, Optimized schemes of enhanced shale gas recovery by CO_2-N_2 mixtures associated with CO_2 sequestration. *Energy Conversion and Management*, 268: 116062.

2022 [J2] **Ma, H.**, Yang, Y.*, Chen, Z.*, Numerical simulation of bitumen recovery via supercritical water injection with in-situ upgrading. *Fuel*, 313: 122708.

2021 [J1] **Ma, H.**, Chen, S., Xue, D., Chen, Y., Chen, Z.*, Outlook for the Coal Industry and New Coal Production Technologies. *Advances in Geo-Energy Research*, 5(2): 119-120.

Conference Proceedings (T=15)

2025 [C15] Ma, H., Zhu Y., Long W., Masnadi M., Heath G., Balcombe P., Ravikumar A. P., Measurement-informed, geospatial, life cycle greenhouse gas emissions of global liquefied natural gas supply chains, *ACS Spring 2025*, San Diego, CA, USA, March 23-27.

2024 [C14] Xue, Z., **Ma, H.**, Chen, Z., Combining Geothermal Energy Recovery and Carbon Sequestration in EGS Using Different Working Fluids, *American Geophysical Union Annual Meeting (AGU24),* Washington, D.C., USA, December 9-13

2024 [C13] **Ma, H.**, Xue, Z., Yang, Y., Chen, Z., System Assessment of CO_2-Enhanced Coalbed Methane Production in Mannville Canada, *American Geophysical Union Annual Meeting (AGU24)*, Washington, D.C., USA, December 9-13

2024	[C12] **Ma, H.**, Tullos, E., Lyon, D., Gupta, S., Ravikumar, A. P., Discrepancy in Intermittency Accounting Methods of Methane Emissions from Aerial-Based Surveys of Oil and Gas Facilities, *American Geophysical Union Annual Meeting (AGU24)*, Washington, D.C., USA, December 9-13
2024	[C11] **Ma, H.**, Xue, Z., Yang, Y., Chen, Z., Environmental Cost of large-scale underground hydrogen storage in Saline Aquifers, *American Geophysical Union Annual Meeting (AGU24)*, Washington, D.C., USA, December 9-13
2024	[C10] **Ma, H.**, Zhu, Y., Long, W., Masnadi, M., Heath, G., Balcombe, P. Ravikumar, A. P., Measurement-informed, Geospatial, Life-cycle Greenhouse Gas emissions of Global Liquefied Natural Gas Supply Chains. *Energy Emissions Modeling & Data Lab 2024 Annual Event,* Austin, USA, October 22-23
2024	[C9] **Ma, H.**, Chen, Z., & McCoy, S. T. Motivating CO_2-Enhanced Oil Recovery to Geologic Carbon Storage by Integrating Techno-Economic and Life-Cycle Assessments. *17th International Conference on Greenhouse Gas Control Technologies (GHGT-17)*, Calgary, Canada, October 20-24
2024	[C8] **Ma, H.**, An, K., Nabil, S. K., Kibria, M. G., & McCoy, S. T. Uncertainty Analysis on CO_2-Ethanol-Synthetic Aviation Fuel Pathways. *17th International Conference on Greenhouse Gas Control Technologies (GHGT-17)*, Calgary, Canada, October 20-24
2024	[C7] Xue, Z., **Ma, H.**, Chen, Z., Integration of geothermal energy recovery and carbon sequestration of an EGS by CO_2-water mixtures, *SPE Western Regional Meeting*, Palo Alto, CA, USA, April 16-18
2023	[C6] **Ma, H.**, Yang, Y., Xue, Z., Chen, Z., Techno-economic feasibility of CO_2 storage associated with enhanced coalbed methane production: a case study of Manville coal. *Canadian Chemical Engineering Conference (CSChE 2023)*, Calgary, Alberta, Canada, October 29 – November 1
2023	[C5] Song, Y., Song, Z., Meng, Y., **Ma, H.**, Chen, Z., Zhang, Y., Zhang, L., Feng, D. Effect of CO_2 dissolution on fluid flow and phase behavior in oil-water systems in shale reservoirs. URTEC-3860690-MS. *SPE/AAPG/SEG Unconventional Resources Technology Conference (URTeC)*, Denver, CO, USA, June 13-15
2023	[C4] **Ma, H.**, Yang, Y., Xue, Z., Chen, Z.*, Comparative machine learning frameworks for forecasting CO_2/CH_4 competitive adsorption ratio in shale. SPE-212994-MS, *SPE Western Regional Meeting*, Anchorage, AK, USA, May 22-25
2023	[C3] Xue, Z., **Ma, H.**, Chen, Z.*, Numerical investigation of energy production from an enhanced geothermal system associated with CO_2 geological sequestration. SPE-213041-MS, *SPE Western Regional Meeting*, Anchorage, AK, USA, May 22-25
2022	[C2] **Ma, H.**, McCoy, S. T.*, Chen, Z.*, Economic and engineering co-optimization of CO_2 storage and oil recovery, *16th Greenhouse Gas Control Technologies Conference (GHGT-16)*, Lyon, France, October 23-27

2022 [C1] Liu, H.*, Cui, L., Liu, Z., Zhou, C., Yao, M., **Ma, H.**, Liu, Q. Using machine learning method to optimize well stimulation design in heterogeneous naturally fractured tight reservoirs. SPE-208971-MS, *SPE Canadian Energy Technology Conference*, Calgary, Alberta, Canada, March 16-17

Preprints & In-Progress (T=7)

Long, W., Jing, L., Liu, Z., Jabber, M., Chen, Z., Moya, D., Ren, B., **Ma, H.**, Littlefield, J., Ramadan, F., Qahtani, A., Guan, D., Bi, X., Bergerson, J. A., Ravikumar, A. P., Houjeiri, H., Masnadi, M. S.*, Global Carbon Intensity of Liquefied Natural Gas Value Chain (in review, *Nature*)

Ma, H., Nabil, S. K., An, K.*, Nishikawa, E., Kibria, M. G., Bergerson, J. A., Chen, Z., McCoy, S. T.*, One- or two-step processes: which have a lower GHG impact for production of synthetic aviation fuel via indirect CO_2 electrolysis? (in review, *Carbon Capture Science & Technology*)

Ma, H., Kou, Z., Lu, Y.*, Xue, Z.*, Zhang, K., Deng, P., Di, C., Chen, Z., The Role of Cybersecurity and Cyber-Attacks in the Natural Gas and Hydrogen Industry, (in review, *Gas Science and Engineering*)

Ma, H.*, Yang, Y., Xue, Z., Long W., Zhu Y., Chen, Z., Towards Large-scale Underground Hydrogen Storage: Business Model Insights Through Holistic System Analysis, (to be submitted to *Nature Communication*, July 2025)

Ma, H., Zhu, Y., Long, W., Masnadi, M., Heath, G., Balcombe, P. Ravikumar, A. P.*, Geospatial Climate Change Insights on Global Liquefied Natural Gas Supply Chains: A Measurement-informed Life-cycle Analysis, (to be submitted to *Nature Climate Change* June 2025).

Ma, H., McCoy, S. T.*, Chen, Z.*, Economic and environmental trade-offs of CO_2 enhanced oil recovery associated with carbon sequestration: an integrated techno-economic and life-cycle assessment framework of Weyburn oilfield, (to be submitted to *Environmental Science & Technology,* May 2025).

Ma, H., Tullos, E., Lyon, D., Gupta, S., Ravikumar, A. P.*, Reconciling the Impact of Intermittency of Aerial-based Methane Measurement Surveys, (to be submitted to *Environmental Science & Technology* June 2025)

Invited Talks

2022 Unconventional Resources Research Seminar Series, China University of Petroleum (East China), Virtual, November 23.

2022 CO_2 Capture and Utilization Workshop, jointly University of Calgary and University of Toronto, October 13.

2022 Multi-group research seminars, jointly ETH Zürich, University of Pennsylvania, Delft University of Technology, and University of Calgary, Virtual, August 15.

LEADERSHIP AND PROFESSIONAL SERVICE

Conference Technical Committees

2025 Disciplinary of Energy Management, *SPE Annual Technical Conference and Exhibition (ATCE)*, Houston, USA, October 20–22.

Conference Session Chairs

2025 **Ma, H.**, Capello, M. A., Risk Management and Sustainable Solutions in Energy Sector: Bridging Technology, Business, and Resilience, *SPE Annual Technical Conference and Exhibition (ATCE)*, Houston, USA, October 22.

2025 **Ma, H.**, Long, W., Ravikumar, A. P., Masnadi, M. S., Advances in Translating Measured Methane Emissions into Greenhouse Gas Inventories for the Oil and Gas Supply Chains, *AGU Annual Meeting (AGU25)*, New Orleans, LA, USA, December 15-19.

2024 **Ma, H**., Ravikumar, A., Sherwin, E., Johnson, M., Advances in Methane Emissions Reconciliation and Uncertainty Estimation Across Oil and Gas Supply Chains, *AGU Annual Meeting (AGU24)*, Washington D.C., USA, December 9-13.

2024 Liu, S., Neil, C., **Ma, H.**, Principles of Geologic Storage: Geochemical Interactions, Geomechanics, Hydrodynamics, and Caprock Integrity, *AGU Annual Meeting (AGU24)*, Washington D.C., USA, December 9-13.

To Academia

Reviewer, Journals (~130 to date): *Renewable & Sustainable Energy Reviews, Applied Energy, Energy Conversion and Management, Energy, Energy Strategy Reviews, Energy Reports, Geoenergy Science and Engineering; International Journal of Greenhouse Gas Control*

Guest Editor, Special Issue on Gas Geo-storage and Unconventional Energy Sources Recovery, *International Journal of Coal Science & Technology*, 2024

Conference Manuscript Reviewer, 58th US rock mechanics/geomechanics symposium, *American Rock Mechanics Association (ARMA)*, 2024

Conference Manuscript Reviewer, *SPE Annual Technical Conference and Exhibition (ATCE)*, 2025

To University

Mentor, International Student Mentorship Program, University of Calgary, 2023

Sponsor Representative, CPE Grad Student Association, University of Calgary, 2021-2022

Vice-President, Chinese Association of EMS College, Penn State University, 2017-2018

To Community

Judge, 2024 SPE Student Chapter Awards 2024

Associate Director, Chinese-Canada Petroleum Society, 2023-2024

Co-Chairperson, SPE North America Students Symposium, 2023

Team Lead, SPE Calgary Section Growth & Development Group, 2022-2023

Volunteer, Startup Village Competition for 2023 ATCE, 2023

CERTIFICATES

University Teaching and Learning, Taylor Institute, University of Calgary, 2022

IBM Artificial Intelligence Engineering, Coursera, 2021

PROFESSIONAL ASSOCIATIONS

Society of Petroleum Engineers (SPE)
American Chemical Society (ACS)
American Geophysical Union (AGU)